ELE 458 Lab #5: Digital Control of Coupled Water Tanks

1 Introduction

The coupled-tank system in the URI Controls Laboratory is a scale model of process control systems developed by chemical engineers. The laboratory system (see Fig. 1) consists of a basin of water, a pump which sends water to an upper tank, and a lower tank. Water flows through an orifice at the bottom of the upper tank into the lower tank, and through an orifice at the bottom of the lower tank into the basin. The task is to control the water level in the lower tank to achieve a specified level.

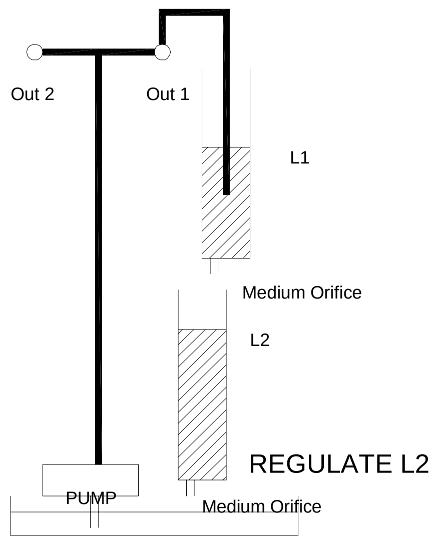

Figure 1: Schematic of a two-tank system.

The mathematical description of the coupled-tank system is given by a 2nd-order nonlinear state-space model. The state variables are

$$x_1 = \text{water level in tank 1 cm (shown as L1 in Fig. 1)}$$
$$x_2 = \text{water level in tank 2 cm (shown as L2 in Fig. 1).}$$

The state-space model is

$$\begin{aligned} \dot{x_1} &= -\alpha_1\sqrt{2gx_1} + \beta u \\ \dot{x_2} &= \alpha_1\sqrt{2gx_1} - \alpha_2\sqrt{2gx_2}, \end{aligned} \tag{1}$$

where g is the acceleration due to gravity and u is the voltage applied to the pump. The pump input u should be between 0 and 30 volts. The numerical values of the parameters are

$$\begin{aligned} g &= 981\ cm/sec^2 \\ \alpha_1 &= 7.84\text{e-}3 \\ \alpha_2 &= 9.09\text{e-}3 \\ \beta &= 0.225. \end{aligned}$$

In order to design a linear control system, the nonlinear state-space model must be linearized about an equilibrium point. The linear control system is then used to control deviations from this equilibrium point. The tanks are each 30 cm tall, so we choose an equilibrium state in which the upper tank is half full. The lower tank will then have an equilibrium height that can be calculated. Let the vector of equilibrium heights be

$$\mathbf{x}_e = \begin{bmatrix} \mathbf{x}_e(1) \\ \mathbf{x}_e(2) \end{bmatrix} = \begin{bmatrix} 15 \\ \mathbf{x}_e(2) \end{bmatrix}.$$

Plugging this constant vector into (1) with $u(t) = u_e$, an unknown constant equilibrium value, and remembering that the derivative of a constant equals zero, yields

$$\begin{aligned} 0 &= -\alpha_1\sqrt{2g\mathbf{x}_e(1)} + \beta u_e \\ 0 &= \alpha_1\sqrt{2g\mathbf{x}_e(1)} - \alpha_2\sqrt{2g\mathbf{x}_e(2)} \end{aligned} \tag{2}$$

The first of these equations can be solved for the equilibrium value of the plant input:

$$u_e = \frac{\alpha_1\sqrt{2g\mathbf{x}_e(1)}}{\beta}. \tag{3}$$

The second equation can be solved for the equilibrium height in tank 2:

$$\mathbf{x}_e(2) = \frac{\alpha_1^2\mathbf{x}_e(1)}{\alpha_2^2}. \tag{4}$$

A linear control system can be used to control deviations from the equilibrium point. In order to obtain a linear state-space model for the deviations, let

$$\mathbf{x} = \mathbf{x}_e + \mathbf{w}, \text{ or } x_1 = \mathbf{x}_e(1) + w_1,\ x_2 = \mathbf{x}_e(2) + w_2, \tag{5}$$

where $\mathbf{w}$ is a vector of deviations from the equilibrium tank water levels, and let

$$u = u_e + v, \tag{6}$$

where v is the deviation of the control input from its equilibrium value. Substituting (5) and (6) into (1) yields

$$\begin{aligned} \dot{w}_1 &= -\alpha\sqrt{2gx_1} + \beta(u_e + v) \\ \dot{w}_2 &= \alpha\sqrt{2gx_1} - \alpha\sqrt{2gx_2} \end{aligned} \tag{7}$$

Consider the Taylor series expansion of a function $f(x)$ about the point x_0:

$$f(x) \approx f(x_0) + \frac{df}{dx}|_{x=x_0}(x - x_0).$$

This Taylor series approximation can be used to obtain a linear approximation to the square-root functions in (7). For example,

$$(2gx_1)^{\frac{1}{2}} \approx \sqrt{2g\mathbf{x}_e(1)} + \sqrt{\frac{g}{2\mathbf{x}_e(1)}}w_1.$$

Substituting this approximation into (7) yields

$$\begin{aligned} \dot{w}_1 &= -\alpha_1\sqrt{2g\mathbf{x}_e(1)} - \alpha_1\sqrt{\frac{g}{2\mathbf{x}_e(1)}}w_1 + \beta u_e + \beta v \\ &= -\alpha_1\sqrt{\frac{g}{2\mathbf{x}_e(1)}}w_1 + \beta v \text{ (see (3) to see why terms cancel)} \\ \dot{w}_2 &= \alpha_1\sqrt{2g\mathbf{x}_e(1)} + \alpha_1\sqrt{\frac{g}{2\mathbf{x}_e(1)}}w_1 - (\alpha_2\sqrt{2\mathbf{x}_e(2)g} + \alpha_2\sqrt{\frac{g}{2\mathbf{x}_e(2)}}w_2) \\ &= \alpha_1\sqrt{\frac{g}{2\mathbf{x}_e(1)}}w_1 - \alpha_2\sqrt{\frac{g}{2\mathbf{x}_e(2)}}w_2 \text{ (see (4) to see why terms cancel).} \end{aligned} \tag{8}$$

Thus, the linearized state-space model that describes deviations from the nominal is

$$\begin{bmatrix} \dot{w}_1 \\ \dot{w}_2 \end{bmatrix} = \begin{bmatrix} -\gamma_1 & 0 \\ \gamma_1 & -\gamma_2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} + \begin{bmatrix} \beta \\ 0 \end{bmatrix} v, \tag{9}$$

where

$$\gamma_1 = \alpha_1 \sqrt{\frac{g}{2\mathbf{x}_e(1)}}, \ \gamma_2 = \alpha_2 \sqrt{\frac{g}{2\mathbf{x}_e(2)}}$$

The interconnection of the linear control system and the nonlinear plant is shown in Fig. 2. The system from v to $\mathbf{w}$ is nonlinear, but it is described approximately by the linear system equations in (9). This linear system is the one used in developing the linear control algorithm.

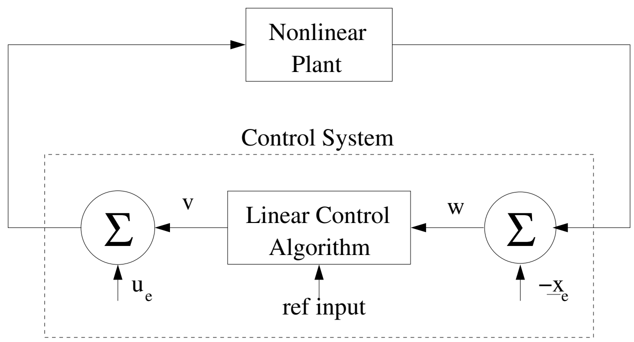

Figure 2: The nonlinear plant connected to a linear control algorithm. The equilibrium state vector is subtracted from the measured state vector, and the equilibrium plant input value is added to the output of the linear control algorithm. The result is that the linear control algorithm is controlling the deviations from equilibrium.

2 Linear State-Feedback Tracking System

The purpose of this lab is to design a linear state-feedback tracking system to enable the coupled tank system to follow a step command. The desired settling time is $T_s = 20$ sec. A tracking system for the linear plant model is shown in Fig. 3. The same linear tracking system can be

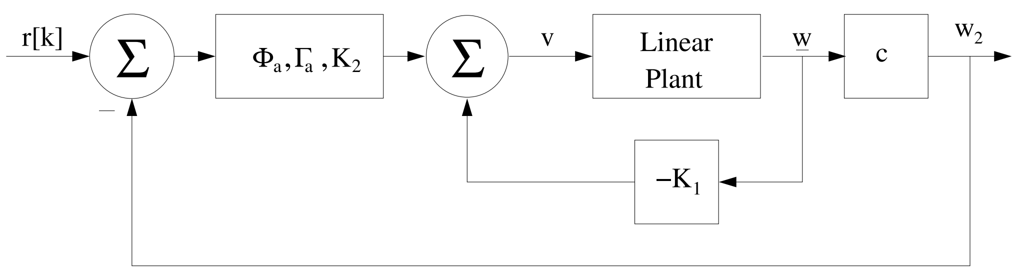

Figure 3: Tracking system for the linear plant model. The reference input is specifying the level in tank 2 as a deviation from the equilibrium value of $\mathbf{x}_e(2)$ cm.

connected to the nonlinear plant, as shown in Fig. 4.

In order to simulate the response of the control system shown in Fig 4, use three step input blocks to create the reference input signal as follows:

$$r(t)] = \begin{cases} 12, 0 \leq t < 45 \\ 15, 45 \leq t < 90 \\ 12, 90 \leq t < 135. \end{cases} \tag{10}$$

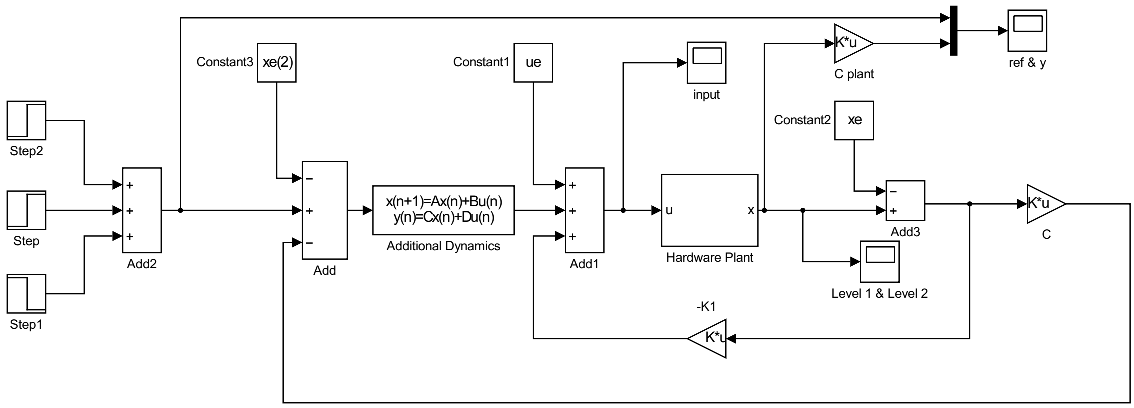

Figure 4: Tracking system for the actual nonlinear plant. The reference input is commanding the level in tank 2. In order to connect the linear control system to the nonlinear plant, the equilibrium state vector and plant input must be subtracted and added, respectively, as shown in Fig. 2.

From the block diagram in Fig. 4, it can be seen that the initial value of the plant input is

$$u[0] = u_e + \mathbf{K}_2\mathbf{x}_a[0] - \mathbf{K}_1\mathbf{w}[0]. \tag{11}$$

The system starts with both tanks empty, so $\mathbf{w}[0] = -\mathbf{x}_e$, and the initial value of the plant input is

$$u[0] = u_e + \mathbf{K}_2\mathbf{x}_a[0] + \mathbf{K}_1\mathbf{x}_e. \tag{12}$$

In order to insure a smooth start-up, the initial plant input is set to zero by properly initializing the state variable of the additional dynamics block:

$$\mathbf{x}_a[0] = -(u_e + \mathbf{K}_1\mathbf{x}_e)/K_2. \tag{13}$$

3 Actual Control System

Because the additional dynamics are designed to track a constant reference signal, the control system will also reject constant disturbance signals anywhere in the feedback loop. Thus, the three constant blocks containing the equilibrium values of the state and input can be removed. The resulting control system is shown in Fig. 5.

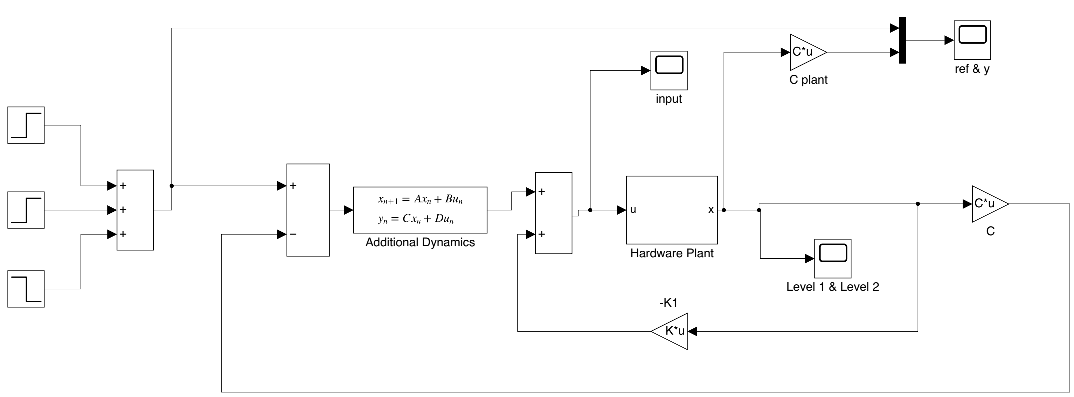

Figure 5: Tracking system that is equivalent to the one shown in Fig. 4. The initial state of the additional dynamics block here is `xa0=0`.

3.1 Specific Tasks

1. Run the m-file, given in the next section, to design the control system.

2. Explain the following to the course instructor.

   (a) Choice of sampling interval

   (b) Choice of pole locations

   (c) What reference command signal is created by the three step blocks in Fig. 5 (see equation (10))?

3. Download and run the Simulink model for the system shown in Fig. 5. Look at the `ref & y` scope (yellow is the reference signal, blue is the measured height in tank 2). Double click on the `Hardware Plant` block. You will see the Simulink blocks used to simulate the nonlinear plant described by equation (1).

4. Change the height of the top step block to 16 and the height of the bottom step block to -7. This corresponds to commands of 16, 19, and 12 cm. Run the simulation and notice the following: **the plant input exceeds 30 volts** early in the run, the **height in tank 1 exceeds 30 cm** (in the hardware system the water would overflow, gushing out the top of the upper tank!), and at around 90 seconds **the plant input goes negative** (the real pump cannot respond to a negative input - it cannot suck water out of the tank). All three of these problems can be addressed by using a "smart integrator" for the additional dynamics (Section 8.5 in the book introduces this topic but more complicated code is needed for the coupled water tanks).

4 Matlab Design

```
load sroots
noise=0; % set noise=1 to simulate noisy sensor measurements
alpha1=7.84e-3; % coefficients of the nonlinear state-space model
alpha2=9.09e-3;
beta=0.225;
g=981;
xe1=15; % choose equilibrium height of tank 1 to be 15 cm
ue=alpha1*sqrt(2*g*xe1)/beta; % calculate equilibrium plant input using equation (3)
xe2=alpha1^2/alpha2^2*xe1; % calculate equilibrium height of tank 2 using equation (4)
xe=[xe1;xe2];
f1=alpha1*sqrt(g/(2*xe1));
f2=alpha2*sqrt(g/(2*xe2));
A=[-f1 0;f1 -f2]; % linearized state-space model
B=[beta;0];
C=[0 1];
Ts=20;
T=0.1;
[phi,gamma]=c2d(A,B,T);
spoles=s3/Ts;
zpoles=exp(T*spoles);
phia=1;
gammaa=1;
[K1,K2,delta1,delta2]=tsd(phi,gamma,C,phia,gammaa,zpoles,T,'place')
dsm_tssf(phi,gamma,C,phia,gammaa,K1,K2) % calculate stability margins
```